

STEPHEN SOMMER

Software Developer | Machine Learning Graduate Student
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EDUCATION

Georgia Institute of Technology <i>M.S. Computer Science, Machine Learning Specialization</i>	Atlanta, GA Aug. 2025 – May 2027
Winona State University <i>B.S. Computer Science, Minor in Psychology, GPA 3.86</i>	Winona, MN Aug. 2021 – May 2024
Rochester Community and Technical College <i>A.S. Computer Science, GPA 4.0</i>	Rochester, MN Aug. 2017 – May 2021

EXPERIENCE

Developer <i>Fastenal</i>	Winona, MN Jan. 2023 – Present
- Implemented content filtering and a data ingestion pipeline for an enterprise AI copilot using Azure OpenAI and Azure AI Foundry with SharePoint integration, enabling automated product identification from images with confidence scoring and Excel-based reporting. - Built and maintained CI/CD pipelines in Azure DevOps for ML workflows, integrating Git version control, model versioning, Terraform-provisioned infrastructure, and automated deployment. - Designed enterprise BI solutions using Power BI and Microsoft Fabric, including dashboards, semantic models, data marts, and KPI documentation to support analytics and reporting needs. - Built and maintained ETL pipelines in Oracle PL/SQL and Python on Databricks, feeding the enterprise data warehouse and data lake, producing curated datasets and owning source-to-report data quality, cleaning, and validation for downstream analysts and reporting teams.	
Software Developer <i>Winona State University</i>	Winona, MN Aug. 2022 – May 2024
- Developed and maintained web applications for Winona State University main website and the Computer Science's department website at Winona State, enhancing user experience and site functionality. - Configure and deployed virtual machines for academic courses, providing students with consistent development environments and technical infrastructure. - Conducted machine learning research on cybersecurity (R2L intrusion detection) and medical imaging (prostate MRI segmentation), improving detection accuracy by 10% through ensemble methods and deep learning techniques.	
Additional Professional Experience <i>Various (Mayo Clinic, Watlow, Benchmark)</i>	Rochester/Winona, MN May 2008 – Jan 2023
- Developed problem-solving and precision skills in clinical IT and manufacturing, supporting complex system troubleshooting. - Diagnosed and repaired hardware and electronics in regulated environments, including desktop/laptop support and patching at Mayo Clinic, and equipment setup, calibration, and test operations under strict quality, ESD, and safety standards in manufacturing.	

PROJECTS

liminal-eeg: EEG State Decoding <i>MNE, PyTorch, Deep Learning</i>	GitHub
- Built an EEG preprocessing and feature-extraction pipeline (ICA artifact removal, Welch PSD band power) producing a 4,273-epoch $\times$ 120-feature dataset from 34-subject recordings. - Implementing deep learning models (EEGNet CNN, temporal Transformer with attention) benchmarked against a published SVM baseline under leave-one-subject-out cross-validation.	
Prostate MRI Segmentation <i>Deep Learning, Medical Imaging, PyTorch</i>	GitHub
- Implemented a 3D segmentation pipeline for prostate MRI using nnU-Net v2 on the Medical Segmentation Decathlon dataset, demonstrating applied ML in a clinical imaging domain. - Developed preprocessing, augmentation, and evaluation pipelines suited to low-volume medical imaging datasets.	

Q-learning vs. SARSA on Taxi-v3 | *Reinforcement Learning, Gymnasium, Python*

[GitHub](#)

- Compared off-policy and on-policy tabular TD control under an annealed ϵ -greedy schedule; Q-learning converged to the known-optimal policy (100% success, +7.86 mean return, 13.1 mean steps over 1,000 held-out episodes) while SARSA converged to a more conservative policy at +5.57.
- Built fully seeded, reproducible experiment tooling that regenerates all metrics and figures from a single command.

R2L Intrusion Detection System | *Decision Trees, Python*

[Paper](#)

- Built a decision-tree classifier for remote-to-local (R2L) attack detection on the KDD dataset; published findings on detection performance and class-imbalance challenges in minority attack categories.

TECHNICAL SKILLS

Programming Languages: Python, SQL (Oracle PL/SQL, PostgreSQL, T-SQL), R, C#, Java, JavaScript

AI/ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Azure OpenAI, Azure AI Foundry, Azure ML, NLP, Computer Vision, Reinforcement Learning

Data Engineering & Warehousing: ETL, Data Warehousing, Data Modeling, Data Curation, Data Quality, Microsoft Fabric, Databricks, Apache Spark, BigQuery, Toad for Oracle, Jupyter

BI & Analytics: Power BI, DAX, Semantic Models, SSRS, Tableau, Ad Hoc Analysis, Dashboard Design

Cloud & MLOps: Azure (ML, AI Services, Data Lake, DevOps), AWS, GCP, Terraform, Docker, CI/CD, Git

Web Development: React, Flask, FastAPI, .NET, REST APIs, Node.js